

ReactionSeek: LLM-powered literature data mining and knowledge discovery in organic synthesis

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The application of artificial intelligence (AI) to chemical discovery is critically hindered by the inaccessibility of data locked within unstructured scientific literature. Existing data acquisition methods are often manual, limited in scope, or require extensive custom software development, impeding progress in leveraging AI for chemical discovery. Here, we introduce ReactionSeek, a framework that synergistically combines large language models (LLMs) with established cheminformatics tools to automate multi-modal data mining from organic synthesis literature. Through sophisticated prompt engineering with minimal custom code, ReactionSeek extracts and standardizes complex textual, graphical, and semantic chemical information. We validate this framework on the century-spanning Organic Syntheses collection, achieving over 95% precision and recall for key reaction parameters. This enables three applications: the generation of a large, AI-ready dataset; an interactive Synthetic Chatbot (SynChat) for natural language querying of chemical data; and an autonomous analysis that revealed decades-long trends in catalysis. ReactionSeek thus provides a general solution to the data curation bottleneck, representing a step forward in for AI-driven archive mining and knowledge discovery across the chemical sciences.

The integration of artificial intelligence (AI) into the molecular sciences is poised to fundamentally reshape discovery, accelerating innovation from medicine to materials¹. This potential is particularly crucial in organic synthesis, the foundational discipline of molecular construction². While AI-driven strategies have demonstrated notable capabilities in retrosynthetic planning, mechanistic studies, and the discovery of novel reactions³, their application to the frontiers of chemical research is hampered by data accessibility. The vast majority of our chemical knowledge, painstakingly accumulated over a century, remains locked within unstructured literature. This forces a reliance on curated, often proprietary datasets, meaning that current AI models are trained on sanitized information that fails to capture the complexity and diversity of real-world chemistry⁴. Therefore, creating automated and scalable methods to convert this

immense scientific legacy into actionable, machine-readable knowledge is the rate-limiting step toward unlocking genuine, AI-driven discovery in chemistry⁵.

Current approaches to data acquisition each present significant limitations. Commercial databases, while extensive, are often constrained by data inconsistencies, incomplete coverage of specialized reaction types, and restrictive licensing terms. High-throughput experimentation (HTE), on the other hand, generates precise datasets that typically lack chemical diversity and are prohibitively expensive for many research groups^{6,7}. As a result, manual curation from the scientific literature remains a common practice for building domain-specific datasets. This approach is not only labor-intensive and time-consuming but also requires specific chemical expertise and is prone to human error, potentially compromising dataset quality and

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scalability. Therefore, a pressing need exists for an automated, reliable, and scalable methodology to address this data curation bottleneck⁸.

Early efforts to automate chemical data extraction have explored various natural language processing (NLP) techniques⁹. Rule-based and traditional machine learning models struggled with the nuanced language of chemical texts, particularly the complexity of molecular nomenclature, reaction schemes, and the implicit knowledge embedded in experimental descriptions¹⁰. Recently, large language models (LLMs) have demonstrated significant promise due to their advanced

text comprehension abilities^{11–18}. However, compared to data mining efforts in related domains such as materials and biological sciences, mining organic synthesis presents substantial challenges (Fig. 1a). This task requires interpreting not only diverse molecular structures and identifiers but also the complex, context-dependent relationships among them. This goes beyond generic text comprehension; it necessitates a framework that can effectively leverage the LLM's latent knowledge of chemistry and reasoning abilities. While recent studies have utilized LLMs through methods such as fine-tuning^{11,13,15,19,20},

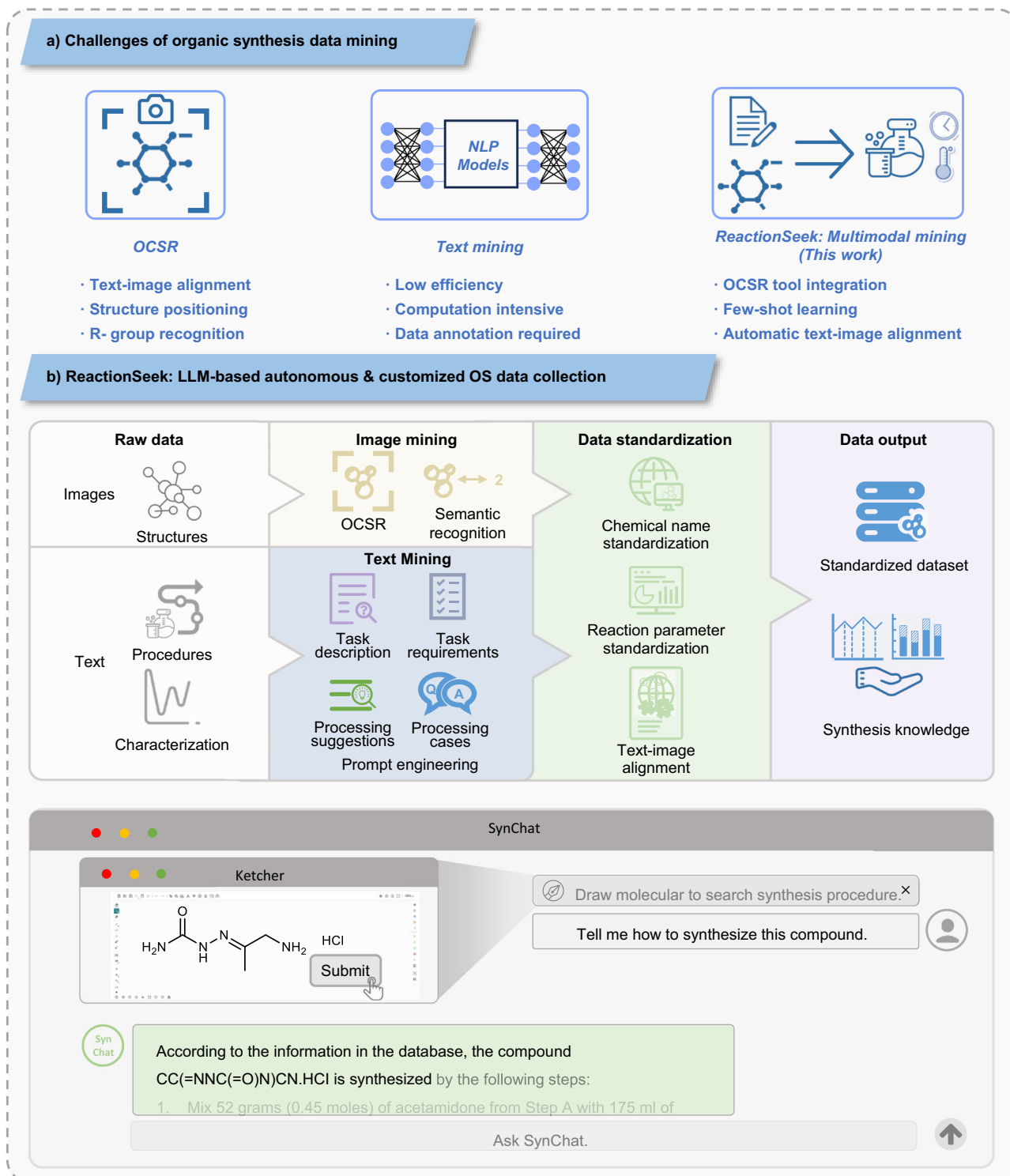


Fig. 1 | Large language models (LLMs) in chemical data mining. **a** Typical challenges for chemical data mining via LLMs and customized tools. **b** Conceptual overview of the ReactionSeek framework.

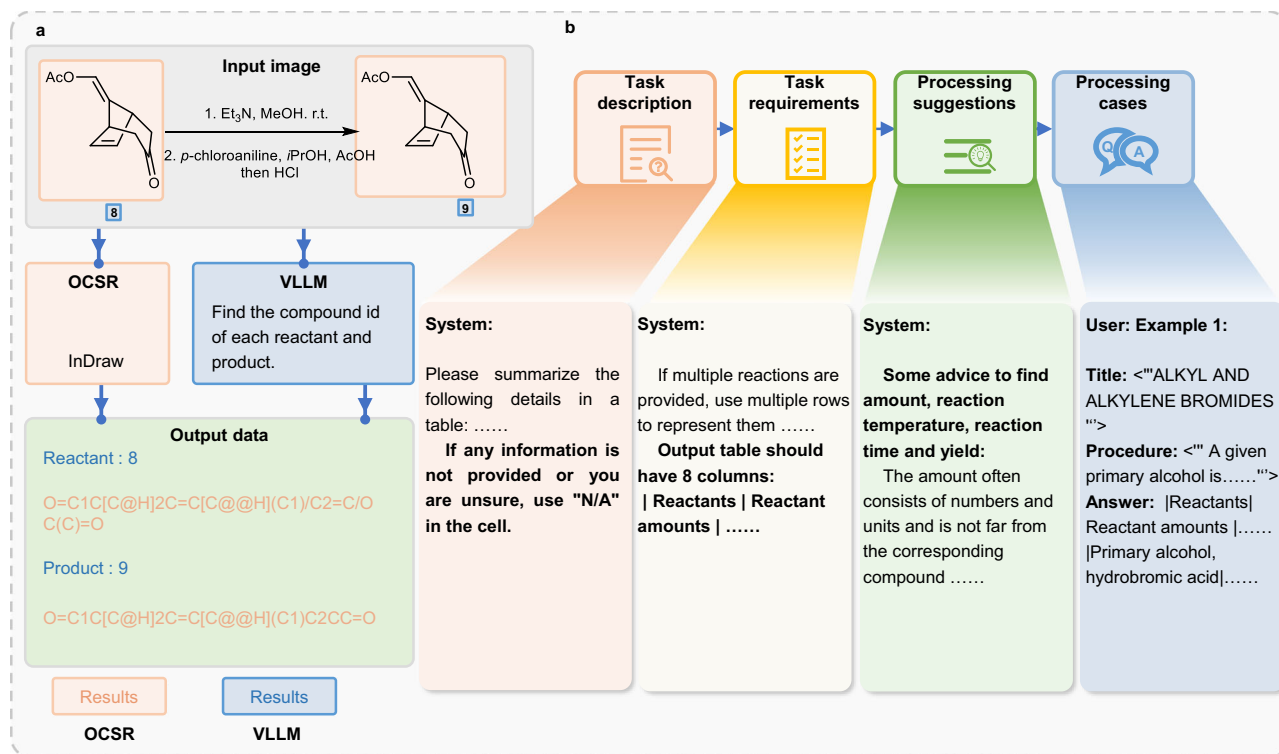


Fig. 2 | Workflow for organic image mining and text mining. **a** Image mining process involves identifying molecular labels within images, assigning them to roles such as reactants or products, and subsequently converting their graphical chemical structures into SMILES using optical chemical structure recognition (OCSR) tools. **b** A complete text mining prompt consists of four parts: task description, task

requirements, processing suggestions, and processing cases. Researchers can freely adjust the prompt content according to the data that needs to be mined, using the framework, and obtain tabulated results through the self-optimization process of the LLM in processing cases.

prompt engineering^{12,16–18,21,22}, and other approaches^{23–28}, these pioneering methods often face limitations. Key issues include a lack of adaptability across different literature styles, overly complex implementations, and the considerable computational cost of model fine-tuning. Additionally, previous LLM-based reaction mining approaches still face several challenges, including: the dispersion of compound information across sub-headings, the conversion of chemical names to SMILES notation, complex compound references, and the identification of compounds in reference images²⁹. Consequently, there remains a persistent need for a more universal and resource-efficient methodology tailored to the intricacies of organic synthesis data mining.

To tackle these challenges, we introduce ReactionSeek. This framework automates the multi-modal extraction of chemical data from scientific literature (Fig. 1b). It employs a hybrid architecture that combines the contextual understanding of LLMs with the chemical precision of established cheminformatics tools. ReactionSeek utilizes a domain-specific prompt engineering strategy, enabling robust and accurate data mining without the need for resource-intensive model fine-tuning. We demonstrate the effectiveness and scalability of this approach by systematically mining the entire century-spanning Organic Syntheses collection³⁰, achieving over 95% precision and recall for key reaction parameters. The resulting AI-ready dataset enabled two downstream applications: the development of SynChat, an interactive chatbot for querying reaction data via natural language, and an autonomous, LLM-driven analysis that successfully identified decades-long historical trends in catalysis. Thus, ReactionSeek provides a solid and transferable solution to the data curation bottleneck, paving the way for large-scale, AI-driven knowledge discovery across the chemical sciences.

Results

The reactionseek data mining framework

ReactionSeek is designed to process chemical literature to extract and organize reaction information, transforming raw textual and graphical data into a curated, machine-readable dataset (Fig. 1b). The workflow comprises three core stages: (1) Image mining for interpreting reaction schemes and structures within figures; (2) Text mining for extracting detailed experimental information from procedural text using prompt-engineered LLMs; and (3) Data standardization for ensuring the accuracy and consistency of the extracted chemical entities and reaction parameters.

Organic Chemistry Image Mining: The accurate interpretation of chemical structures and reaction schemes depicted in images is a foundational challenge in automated chemical data mining. While existing Optical Character Structure Recognition (OCSR) tools can convert molecular structures into machine-readable formats (e.g., SMILES)^{24,25,31–33}, associating these structures with their correct roles (e.g., reactant, product) and textual identifiers within complex figures remains a daunting challenge. ReactionSeek addresses this by employing a multi-modal LLM (GLM-4V) to analyze reaction scheme images (Fig. 2a). The LLM is prompted to identify compound labels within the image and map them to the corresponding molecular structure images, which are subsequently converted to SMILES by OCSR tools (e.g., InDraw³⁴).

To evaluate this image mining module, we tested it on 42 diverse reaction scheme images sourced from the supplementary information (SI) of published literature. The multi-modal LLM (GLM-4V) achieved 91.5% accuracy in correctly identifying and associating the molecular labels and their roles of reactants or products. The InDraw OCSR tool

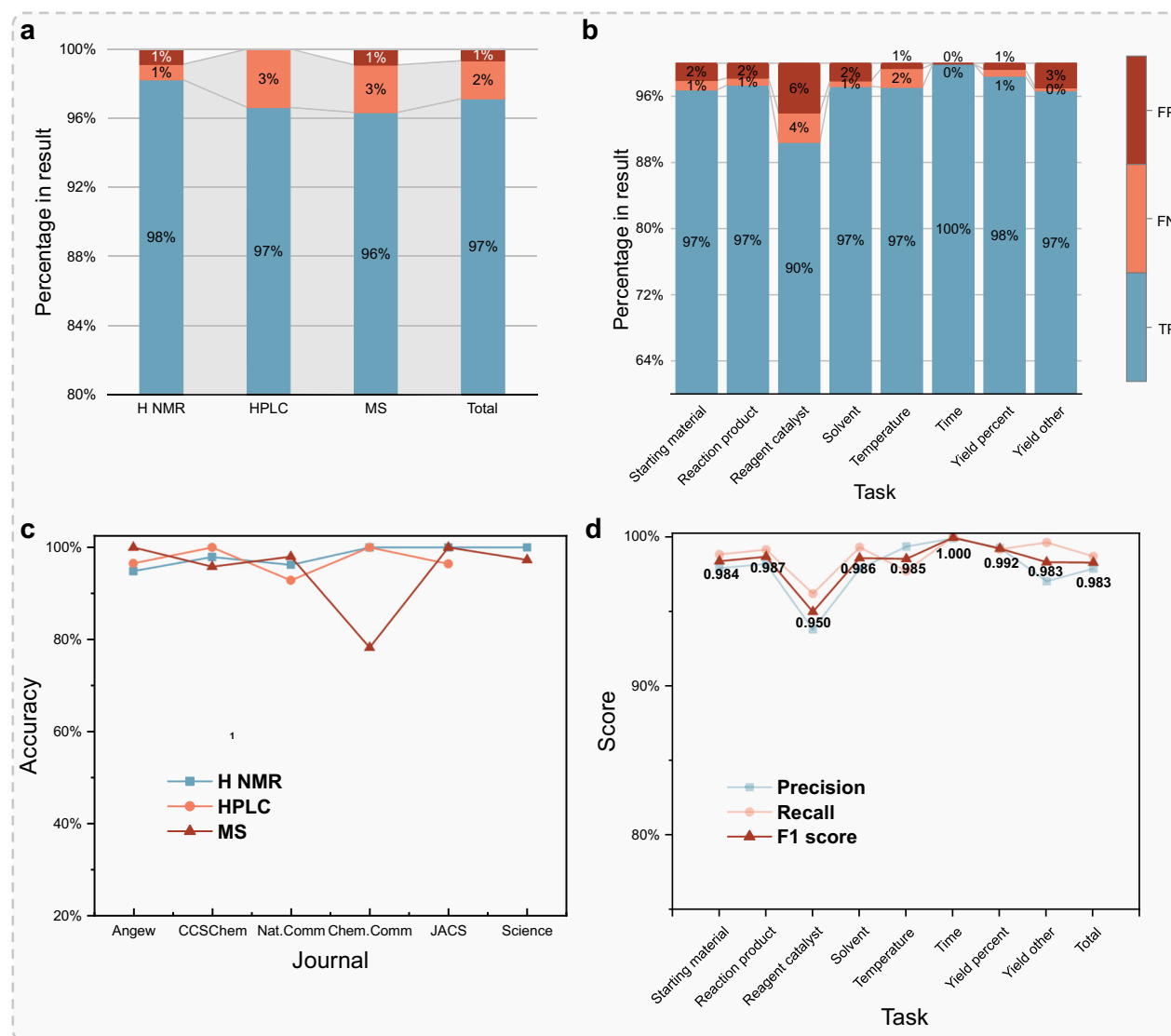


Fig. 3 | Performance of spectroscopic data extraction from the Supporting Information and ChEMU project. a Breakdown of extraction outcomes (TP, TN, FP, FN) for key spectroscopic data types: ¹H NMR, HPLC, and MS. **b** Breakdown of

extraction outcomes for ChEMU 2020 competition as examples. **c** Extraction accuracy for spectroscopic data stratified by source journals. **d** Extraction score for each data type in ChEMU 2020 data and their specific F1 scores.

was used to convert the molecular images to SMILES with 70.1% accuracy (see more details in Supplementary Note 1.1)³⁴. While the overall performance is promising, the accuracy of structure recognition is influenced by the capabilities of the currently integrated OCSR tool, suggesting potential for improvement with more accurate OCSR solutions.

Organic Chemistry Text Mining: Extracting detailed experimental procedures and reaction parameters from unstructured text is central to ReactionSeek. To achieve this with high fidelity and minimal human intervention, we developed a sophisticated prompt engineering pipeline for LLMs (Fig. 2b). This pipeline adapts general-purpose LLMs to the specific nuances of organic chemistry literature, guiding them to extract predefined chemical information and structure into a tabulated format suitable for review and downstream processing. With this pipeline, the LLM can effectively identify compound entities within text segments and establish their complex co-reference relationships. By processing sub-headings and text segments separately, our pipeline successfully achieves the association between compound references in synthetic text segments and the corresponding compounds mentioned in sub-headings.

Our prompt architecture, inspired by previous work on MOF data extraction¹⁷, comprises four key components designed to optimize LLM performance in this domain-specific task (see more details in Supplementary Note 4.1): (1) Task description: This component clearly defines the overall objective of the data mining task and provides fundamental guidelines to the LLM, including instructions to mitigate common issues like output hallucination. (2) Task requirements: Here, we specify the information to be extracted and the output format (e.g., a table with specific columns). This section is strategically placed to reinforce the LLM's adherence to output specifications, countering potential memory decay with long inputs. (3) Processing suggestions: This provides the LLM with expert-derived heuristics in natural language. These suggestions are based on a pre-examination of sample texts by domain experts, who summarize common textual patterns and incorporate essential chemical knowledge to aid the LLM in interpreting chemical narratives. (4) Processing cases: Few-shot examples, consisting of real input texts and their corresponding desired structured outputs, are provided. This allows the LLM to learn the specific input-output patterns and formatting requirements through in-context learning. This structured prompting approach is

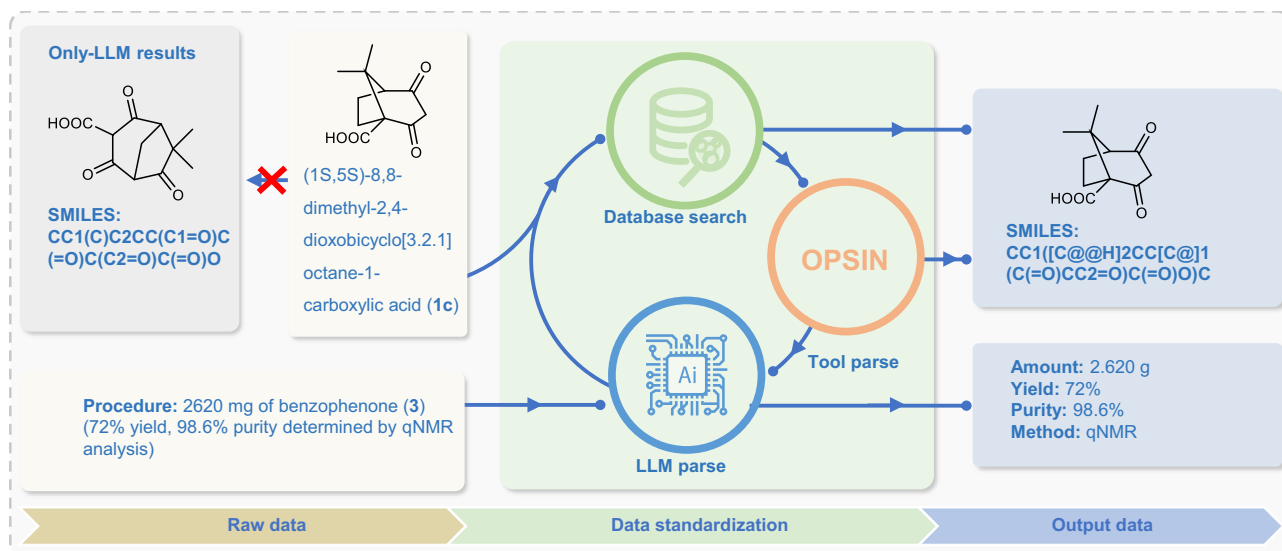


Fig. 4 | Workflow for data standardization. The data standardization process is an iterative loop that synergistically combines database lookups, cheminformatics tool parsing, and intelligent matching via an LLM. The process terminates upon successful identification or when the predefined step limit is exceeded. The power

of this workflow is demonstrated by an example where an LLM, on its own, fails to parse a complex molecular name. In contrast, our workflow correctly identifies the structure and converts it to the canonical SMILES.

designed to ensure the LLM maintains a balance of task memory, semantic comprehension of the chemical context, and accurate execution of the extraction task.

To validate our data mining pipeline, we benchmarked its performance on the authoritative ChEMU^{35–38}. This selection ensures a rigorous comparative analysis, as state-of-the-art models from the ChEMU-2022 competition have explicitly established a high-performance baseline (F1 = 0.968) on this specific dataset. Our pipeline demonstrated a remarkable advantage in the few-shot learning scenario. With only seven annotated examples, it achieved an F1 score of 0.983 in extracting critical reaction components (Fig. 3b), including reactants, products, solvents, temperature, time, and yield (Fig. 3d), representing a further improvement over the previous model (see more details in Supplementary Note 3.3 Table S1).

The text mining efficacy of GLM-4 was also evaluated using a benchmark dataset of 50 reactions obtained from the Supporting Information (SI) of six diverse articles on organic synthesis (see more details in SI References). The extraction results were classified as true positives (TP; correctly identified entities or values), false negatives (FN; omissions), and false positives (FP; incorrect or fabricated information). In synthetic chemistry applications, FPs are typically considered more detrimental than FNs. Our prompting strategy effectively guided the model to extract a variety of experimental data, including compound identifiers, quantities, yields, solvents, reaction conditions, and enantiomeric excess (*e.e.*) values, achieving high precision, recall, and F1 scores for nearly all categories (see more details in Supplementary Note 4.1 Table 2). The only discrepancy was the misidentification of an uncommon solvent mixture or additive (e.g., 18-crown-6), indicating that the model's performance critically relies on its pre-existing chemical knowledge. This limitation can be easily addressed by enhancing the prompt with information about such uncommon entities.

The text mining pipeline was subsequently expanded to extract complex spectroscopic data (NMR, MS, HPLC) by leveraging tailored prompts with GPT-4. We targeted characterization data for 236 molecules from the SI of the same benchmark literature used previously and developed prompts for data mining, including ¹H NMR, MS, HPLC, and IR (see more details in SI 3.1–3.2). The system

demonstrated high efficacy, with extraction accuracies exceeding 95% for most spectroscopic types (Fig. 3a). Notably, despite the inherent complexity of ¹H NMR data, which requires parsing chemical shifts, splitting patterns, and coupling constants, the accuracy of its extraction remained remarkably high. This success suggests that the structured prompts enabled the LLM to effectively extract the in-context chemical knowledge inherent in spectroscopic reports (Fig. 3c). Error analysis highlighted challenges in classifying coupling constants (*J* values) and resolving ambiguities due to inconsistent compound labels. Subsequent analysis indicated that more advanced LLM architectures exhibit greater robustness against such inconsistencies.

Standardization of Extracted Chemical Information: The conversion of diverse compound nomenclature (including IUPAC names, common names, misspelled variants, and chemical formulas) into machine-readable SMILES representations remains a significant challenge in organic reaction mining²⁹. A critical component of ReactionSeek is its data standardization module, which is designed to convert raw extracted text into an accurate, consistent, and AI-ready format (Fig. 4). The module emphasizes two primary curation tasks: name-to-SMILES transformation and unit standardization.

General-purpose LLMs are often unreliable for direct name-to-SMILES transformation, a limitation rooted in their generative instability and lack of domain-specific knowledge training. As reported in previous work³⁹, this deficiency results in a high frequency of critical errors, including unparsable strings, the addition or omission of atoms, incorrect functional group placement, and erroneous backbone structures. Given these significant reliability issues, the direct application of general-purpose LLMs for this conversion task is challenging. ReactionSeek overcomes this by implementing a hybrid workflow: (1) First, unique compound names are compiled from the mining results. (2) Second, these names are processed using established cheminformatics resources, including database lookups (e.g., PubChem⁴⁰) and name-to-structure converters (e.g., OPSIN⁴¹), to generate canonical SMILES. (3) Finally, the LLM, enhanced with vector and fuzzy search capabilities, intelligently maps the mined nomenclature, which is often non-standardized, to the canonical SMILES curated by these specialized tools. This iterative process of data validation, parsing, and LLM-assisted alignment ensures the generation of high-fidelity structural representations (Fig. 4).

Simultaneously, the module leverages the LLM in unit standardization (Fig. 4). Through carefully designed prompts, it standardizes units for all reaction parameters (e.g., time, temperature) to ensure dataset-wide consistency. The final output is a structured CSV file that summarizes each reaction using standardized identifiers and parameters. This curated dataset is not only directly suitable for building predictive models but also serves as the knowledge base for interactive tools such as the SynChat application.

Large-scale reaction mining: a case study of the organic syntheses collection³⁰

To demonstrate the scalability and practical utility of ReactionSeek, we applied it to the systematic mining of the Organic Syntheses collection³⁰, a historically significant and high-quality repository of verified synthetic procedures published since 1921. This collection, comprising 102 volumes as of 2025, represents a valuable resource for studying the evolution of synthetic methodology. For this study, we processed 3103 articles from the first 100 volumes, establishing a comprehensive raw dataset for subsequent analysis and chatbot development (see more details in Supplementary Fig. 4). A benchmark subset of 50 articles from this collection was also curated manually for detailed performance evaluation.

Efficiency of Automated Mining: A key advantage of ReactionSeek is its potential to significantly reduce the time and effort associated with manual data extraction. On the 50-article benchmark set, using GLM-4⁴², the framework required an average of 21.32 s per article for data mining and output formatting. This represents a substantial improvement over manual data mining, which typically requires 5–10 min or longer per article, underscoring the efficiency gains enabled by our LLM-based approach.

Accuracy Assessment on Organic Syntheses³⁰ Benchmark: We evaluated the accuracy of ReactionSeek (with GLM-4) on the 50-article benchmark dataset, employing evaluation criteria tailored to the practical needs of subsequent applications. (1) Reactants and Products: Correct and complete extraction and classification of all compound entities involved as reactants or products were considered critical. (2) Reaction Yield: Precise extraction of yield values. (3) Reaction Scale (Amounts): The focus was on accurately extracting amounts for key reactants and products, with relaxed stringency for ancillary reagents, such as solvents or excess reagents, where order-of-magnitude and proportional relationships are often sufficient. (4) Time and Temperature: Given the potential variability in experimental records for these parameters, the model was assessed on its ability to extract the critical time and temperature values it deemed necessary, rather than achieving absolute accuracy for every recorded value.

As shown in Fig. 5a, GLM-4 demonstrated strong performance in compound identification, particularly for products, with very few FPs in reactant extraction, meeting a key accuracy demand. However, some omissions (FNs) occurred during reactant extraction, likely due to inconsistent textual positioning of reactants. The occurrence of FPs in solvent identification suggested the LLM might rely on its a priori knowledge of common solvents, potentially misclassifying uncommon solvents or reactants.

In numerical data extraction, the model showed high accuracy for reaction yields and product quantities, with no instances of hallucinated yield values on the test set. A notable error pattern emerged in reactant quantity extraction when a compound was added in multiple portions; the model often struggled to accurately aggregate these amounts. This may reflect limitations in the LLM's complex numerical reasoning within linguistic contexts or the absence of specific prompt engineering for such scenarios. Time and temperature extraction, although facing similar complexities, benefited from being listed as separate extraction targets, resulting in comparatively better performance.

Out of the 50 benchmark articles, 48 yielded outputs conforming to the required format after reprocessing of any initially unparseable results. The two articles that failed to produce correctly formatted outputs were significantly longer (9824 and 5502 characters, which may almost exceed the context window of the LLM) and contained more reactions (5 reactions each) than average. In these cases, GLM-4 successfully extracted the information but failed in formatting, suggesting that very long input texts following an already lengthy prompt can lead to the model partially forgetting initial formatting instructions⁴³. Text segmentation is a potential strategy to mitigate this issue.

Comparative Evaluation of LLMs for Text Mining: The performance of LLM-based data mining is significantly influenced by the choice of the underlying model, particularly its parameter size and architecture. To guide model selection for ReactionSeek, we evaluated a diverse range of LLMs on the Organic Syntheses³⁰ benchmark dataset. The models tested included GLM-4, GPT-3.5⁴⁴, GPT-4⁴⁵, Llama⁴⁶, Mixtral⁴⁷, Baichuan⁴⁸, Qwen⁴⁹, and DeepSeek⁵⁰.

Larger models generally exhibited significantly higher performance in extracting chemical information (Fig. 5a, 5d). This suggests that increased parameter size critically impacts an LLM's ability to understand complex chemical contents. Larger models were better at understanding task objectives and correctly identifying relevant chemical entities. Furthermore, the incidence of unparseable or incorrectly formatted responses decreased with increasing model size (Fig. 5b). This improvement is likely due to the enhanced capacity of larger models to handle long token inputs while retaining memory of key prompt instructions, and their superior ability to comprehend complex chemical relationships and knowledge inherent in the context.

Interestingly, while larger models excelled in precision, medium-sized models often achieved comparable recall rates (Fig. 5c). This indicates that once a model possesses a sufficient parameter scale to grasp the fundamental chemical tasks outlined in the prompts, further size increases may offer diminishing profit for simple identification tasks. Persistent error types, such as the aggregation of reactant quantities from multiple additions, were not fully resolved solely by increasing model parameter size, highlighting the need for targeted prompt refinements or algorithmic solutions for such specific challenges (Fig. 5b–d).

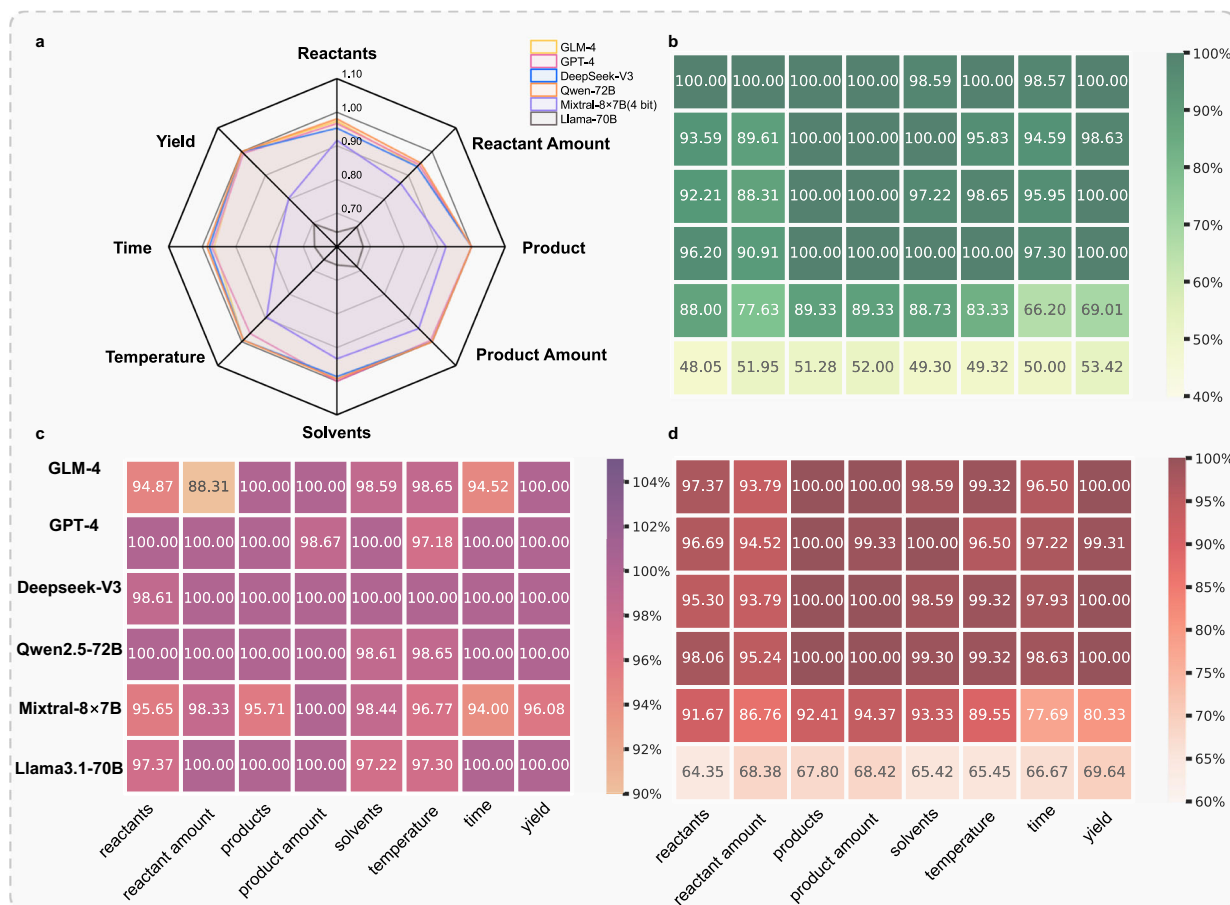
Analysis of successful extractions by small and medium-sized models revealed they typically performed well in articles describing simpler reactions (e.g., fewer compounds, straightforward single-step procedures) and those with shorter text lengths. This suggests that text segment length is a key factor influencing an LLM's ability to generate valid, well-formatted outputs, as shorter texts reduce the cognitive and memory burden on the model. For less complex extraction tasks or when using smaller models, strategies such as text segmentation or employing models specifically adapted for long contexts can improve performance.

Based on the aforementioned evaluations, we ultimately employed GPT-3.5-16k to conduct data mining on Organic Syntheses³⁰ volumes 1–100, spanning from 1921 to 2021. After complete pipeline processing and deduplication, we obtained 5443 unique compounds, 3961 unique reactions, of which 3580 contained yield information.

To demonstrate our pipeline's generalizability to newly published literature, we tested it on all newly published articles (29 articles) from the 2025 volume of Organic Syntheses³⁰ (Fig. 6 and Table 1). Our pipeline correctly extracting most of reaction information from each article. During the standardization phase, with the exception of molecular sieves that cannot be directly represented as SMILES, all other compounds were successfully converted into their correct corresponding SMILES representations. These results confirm our pipeline's robustness, showing its ability to generalize effectively from benchmark datasets to newly published chemical literature.

Table 1 | Reaction mining test results and scores for recently published *Organic Syntheses*³⁰ articles

Evaluation	Reactants	Reactant amount	Products	Product amount	Solvents	Temperature	Time	Yield	Total
TP	171	169	54	60	67	76	75	57	729
FN	2	2	0	2	0	2	1	3	12
FP	1	3	0	0	0	1	1	0	6
Precision	0.9942	0.9883	1.0000	1.0000	1.0000	0.9870	0.9868	1.0000	0.9918
Recall	0.9884	0.9826	1.0000	0.9677	1.0000	0.9744	0.9868	0.9500	0.9838
F1	0.9913	0.9854	1.0000	0.9836	1.0000	0.9806	0.9868	0.9744	0.9878

**Fig. 5 | Comparative performance on the text mining task in the *Organic Syntheses*³⁰ collection text mining task.** The figure displays a comparison various models across four standard metrics: **a** Accuracy, **b** Precision, **c** Recall, and **d** F1 Score.

While modest in size compared to larger reaction databases like USPTO, the value of our minded dataset lies not in its breadth, but in its curated depth. It is composed of the most representative and classical organic reactions across one century, with the exclusive inclusion of gram-scale procedures guaranteeing experimental robustness and reproducibility. Consequently, we propose that this collection can serve as a standardized reference for synthetic chemists and as a high-fidelity benchmark for the development and validation of machine learning models.

LLM-driven discovery of chemical trends from mined data

To demonstrate the potential for data-driven discovery, we utilized ReactionSeek to create a large-scale, structured dataset from the *Organic Syntheses*³⁰ collection. This AI-ready dataset then served as the input for autonomous analysis via Gemini 2.5 Pro⁵¹. By designing specific prompts, we tasked the LLM with uncovering historical trends within this century-spanning compendium, including the emergence

and diversification of asymmetric catalysis and the changing landscape of metal utilization in synthesis (Fig. 7a–d).

In its analysis of asymmetric catalysis (Fig. 7b), the LLM identified a significant increase in such reactions after 1980. It highlighted the pivotal role of transition-metal catalysts, alongside the emergence of organic small-molecule catalysts and enzymatic/biocatalytic systems. The LLM noted the prevalent use of specific metals like Boron, Copper, Titanium, Ruthenium, and Palladium, correlating their use with advancements in well-known methodologies such as CBS reductions, metal-coupling protocols, asymmetric epoxidation (e.g., with Titanium Tartrate or Ti-BINOL systems), asymmetric hydrogenation, cross-coupling reactions, and allylic alkylation. Furthermore, the LLM identified substrate preferences, observing that aldehydes and ketones, as well as halides, were the most dominant, with esters and alkenes also featuring prominently (see Supplementary Note 2.4 for more details).

Regarding metal usage trends (Fig. 7b, 7d), the LLM's analysis revealed an interesting prevalence of tin in specific periods, correctly

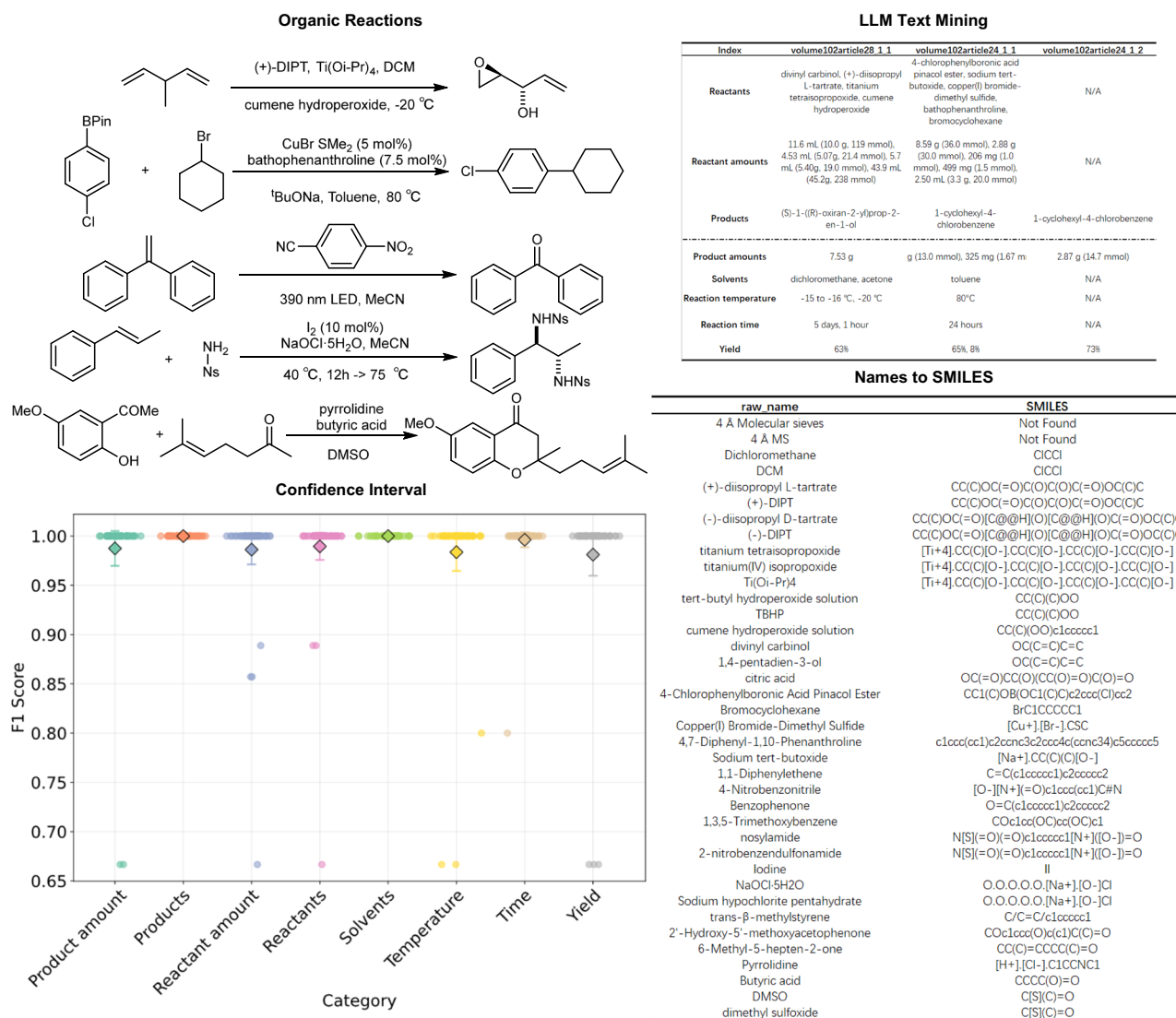


Fig. 6 | Extraction demonstration on the latest article reactions from Organic Syntheses³⁰. Right part shows representative text mining outputs and the corresponding standardized list. Scatter plot showing the distribution of mining F1 scores in selected individual articles. Error bars denote the 95% confidence intervals.

attributing this to the widespread adoption of tributyltin hydride (Bu_3SnH) as a mild radical initiator, reducing agent, and key component in Stille couplings. More broadly, the LLM categorized the evolution of metal usage in Organic Syntheses³⁰ into three distinct epochs: (1) The Main-Group Metal Era (1921–1960): Dominated by Grignard reagents and salt-forming reactions, with less emphasis on metal toxicity. (2) The Incipient Transition (1961–1990): Characterized by the rise of lithium/copper reagents, the early development of palladium catalysis, and a decline in mercury utilization. (3) The Golden Age of Transition-Metal Catalysis (1991–2023): Marked by concentrated growth in Palladium-, Ruthenium-, and Rhodium-mediated transformations.

These LLM-identified trends accurately reflect established paradigm shifts in organic synthesis from stoichiometric to catalytic approaches, from harsh to milder conditions, and from highly toxic reagents towards greener methodologies. This demonstrates that an LLM, when provided with a structured dataset curated by Reaction-Seek, can autonomously develop analytical insights and perform trend analyses consistent with expert chemical knowledge, highlighting the potential for AI-assisted discovery from systematically mined literature.

The synthesis chatbot: interactive access to mined knowledge

To enhance the accessibility and utility of the extensive dataset curated from Organic Syntheses³⁰, we developed SynChat, an interactive tool powered by LLM agents. SynChat allows researchers to query the historical reaction data and associated metadata using natural language, providing a more intuitive and efficient means of accessing specific data compared to traditional search methodology.

The development of the chatbot involved structuring the mined data for effective retrieval by the LLM. Key information from each reaction entry (e.g., title, DOI, reactants, products, SMILES, yield) was compiled and vectorized to a customized vector database⁵². This allows the LLM, when presented with a user query, to perform similarity searches across the Vector Database and retrieve the most relevant supporting texts from the original Organic Syntheses³⁰ (Fig. 8). The LLM agent then synthesizes an answer based on this retrieved information, previous record and reasoning, with prompts designed to ensure strict adherence to the source material, thereby minimizing the likelihood of generating hallucinated or inaccurate responses.

SynChat offers several advantages for chemical researchers: (1) Natural Language Queries with Molecular Structures: It handles complex queries posed in natural language that incorporate molecular

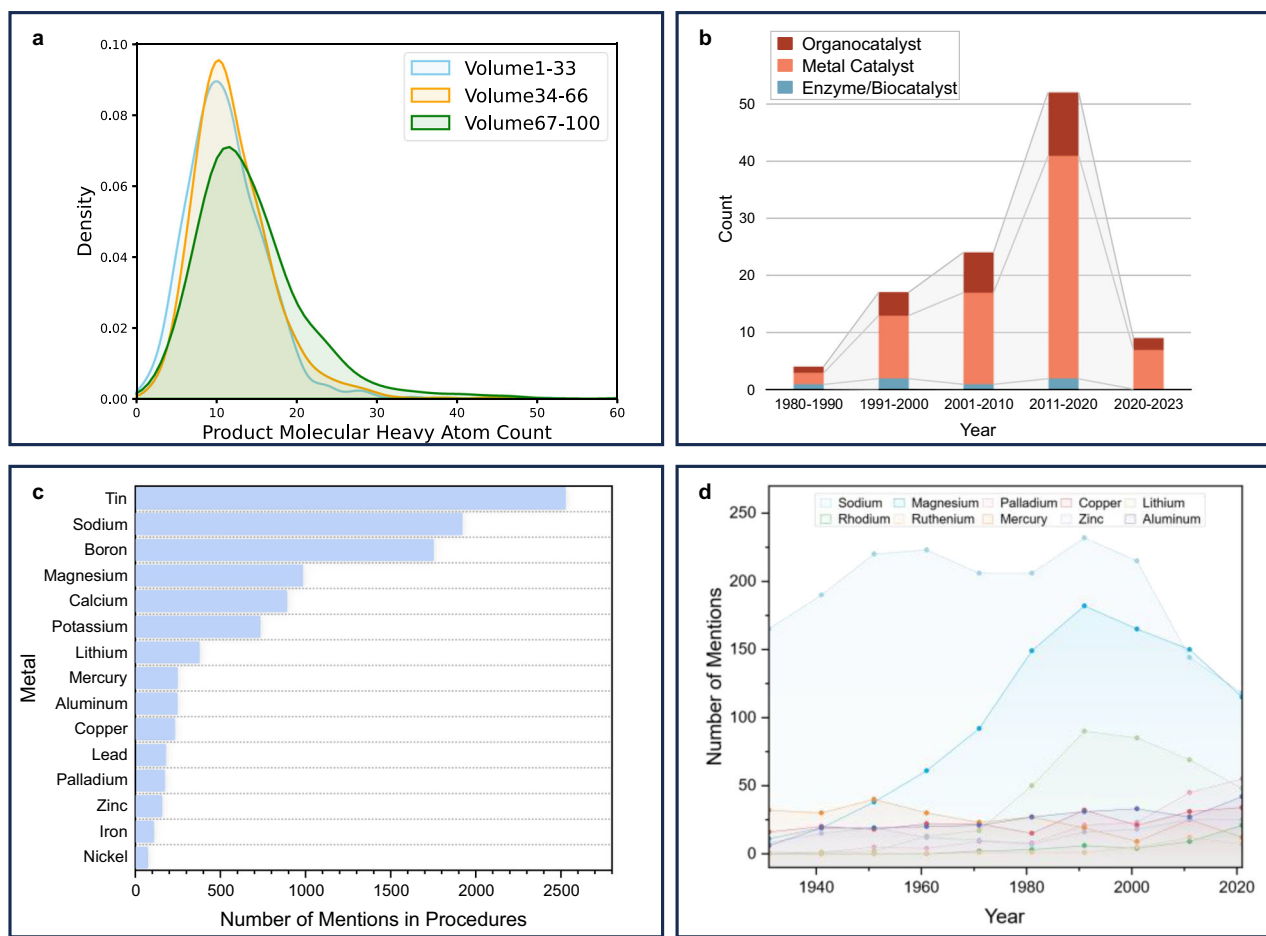


Fig. 7 | Autonomous knowledge discovery from the *Organic Syntheses*³⁰ dataset. The figure showcases analyses performed by Gemini 2.5 Pro on the ReactionSeek generated dataset, which containing 3961 reactions and 5443 compounds spanning from 1921 to 2021. **a** The heavy atom count distribution of products

provides context on molecular complexity. The LLM then identified historical trends in **(b)** asymmetric catalyst usage, **(c)** the utilization of the top 15 metals, and **(d)** the overall evolution of metal types in synthesis over time.

structures or reaction information, increasing the flexibility of data access for users without specialized database query language skills. (2) Enhanced Reliability and Traceability: By citing specific data sources from the *Organic Syntheses*³⁰ collection, the chatbot allows users to consult the original literature for verification and further context easily. (3) Iterative Information Retrieval: The interactive nature of the LLM enables users to refine their queries or ask follow-up questions in multi-turn dialogues, facilitating a more thorough exploration of the available information. This interactive tool exemplifies how LLMs can transform structured and unstructured data into dynamic knowledge resources, bringing AI-powered assistance directly into the research and learning workflows of organic chemists.

Discussion

Here, we present ReactionSeek, an LLM-based framework that effectively automates the extraction of multi-modal data from organic synthesis literature, achieving over 95% accuracy for key reaction parameters. Its primary innovation lies in a sophisticated prompt engineering strategy that confers high accuracy and adaptability across different LLMs without requiring costly fine-tuning. This approach, combined with a hybrid workflow that leverages deterministic cheminformatics tools for SMILES canonicalization, addresses the known unreliability of general-purpose LLMs for chemical tasks. Our analysis thus confirms that intelligent prompt design is the critical factor for successfully parsing the semantic complexities of chemical texts.

ReactionSeek addresses the critical data bottleneck in organic chemistry by offering a scalable method to generate high-quality, AI-ready datasets with minimal manual effort. This approach not only enables autonomous, LLM-driven knowledge discovery, as demonstrated by identifying historical catalysis trends in the *Organic Syntheses*³⁰ collection, but also powers interactive tools like SynChat, which allows researchers to query vast chemical data via natural language. ReactionSeek thus functions not merely as an extraction tool but as a foundational platform for next-generation, AI-driven chemical research.

Current limitations include challenges in (1) handling uncommon chemical names and structures, (2) interpreting complex contextual relationships dispersed across the paper, and (3) a reliance on the performance of the underlying OCSR tool and LLMs. For (1), the structural elucidation of newly synthesized compounds presents significant challenges when these compounds are neither registered in existing databases nor properly systematized through standard nomenclature rules. For (2), for complex compound relationships presented in tables, ReactionSeek still faces challenges in achieving satisfactory mining success rates. For (3), currently available OCSR tools still face significant challenges in R-group identification and complex structure recognition. Concurrently, the developmental stage of LLMs imposes limitations on both the performance of LLMs and existing cheminformatic tools. Future work will, therefore, focus on enhancing robustness through more sophisticated prompt strategies, integrating advanced cheminformatics tools, and expanding the

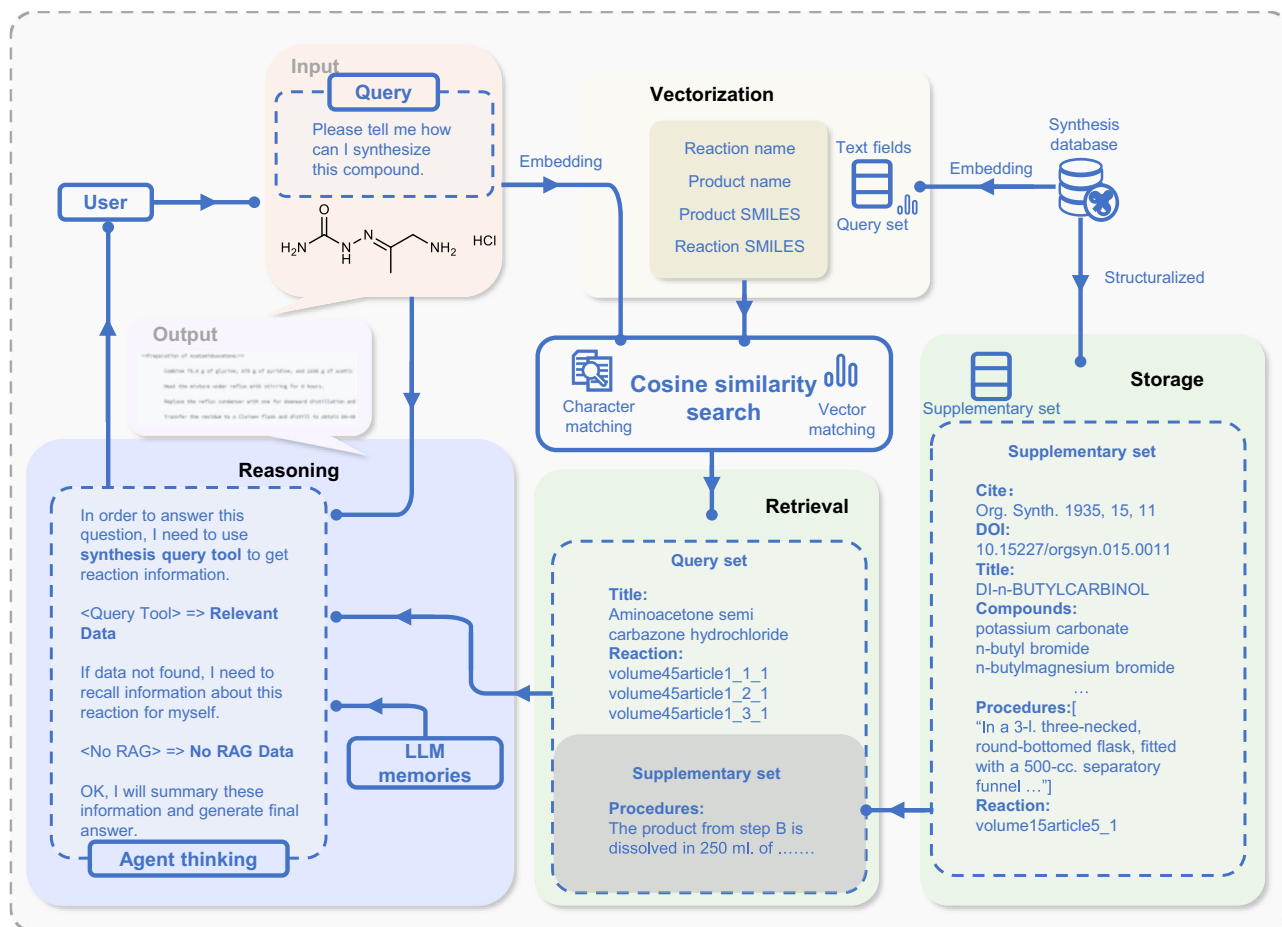


Fig. 8 | SynChat: an AI-powered conversational interface for chemical synthesis knowledge. Retrieval-Augmented Generation (RAG) architecture. The chatbot's backend employs an RAG architecture. A vectorized search first identifies relevant

synthesis documents based on the user's query. This retrieved context is then provided to an LLM Agent, which formulates a comprehensive and accurate response.

framework's scope to capture mechanistic details. Extending ReactionSeek to related fields, such as polymer science or materials science, also presents a promising avenue for broader application.

In summary, ReactionSeek establishes a way for harnessing the vast, unstructured knowledge embedded in scientific literature. By synergistically combining the contextual understanding of LLMs with the rigor of domain-specific tools, our framework provides a robust methodology for accelerating data-driven discovery in chemistry and beyond.

Methods

Prompt design for image mining

Image mining was conducted using the GLM-4V multi-modal large language model. To improve the logical accuracy of the model in determining compound roles, we designed specific instructions requiring the LLM to explicitly analyze the visual orientation of reaction arrows (e.g., distinguishing between reactants and products based on arrow direction). Adopting the structured prompt framework utilized in our text mining workflow, the image mining prompts were specifically tailored to extract the numerical labels associated with reactants, products, and reagents from the reaction schemes. Following role identification, chemical structure recognition was performed using InDraw³⁴, which automatically segmented and identified each compound structure before converting them into canonical SMILES strings for storage. Full details regarding the prompt engineering strategies and templates are provided in Supplementary Note 1.1.

Prompt design for text mining

To extract detailed experimental procedures and reaction parameters from unstructured organic chemistry literature with high fidelity, we developed a structured prompt engineering pipeline tailored for Large Language Models (LLMs). This pipeline was designed to guide the LLM in identifying compound entities, resolving co-reference relationships, and structuring extracted data into a standardized tabular format. The prompt architecture, inspired by methodologies for metal-organic framework (MOF) data extraction¹⁷, consists of four strategic components optimized for chemical text analysis:

1. **Task Description:** This module defines the global objective of the data mining task. It includes explicit constraints and negative constraints to mitigate common LLM issues, such as hallucination, ensuring the model focuses strictly on the provided text.
2. **Task Requirements:** To ensure output consistency, this module specifies the exact information fields to be extracted and enforces a strict tabular output format. This instruction block is strategically positioned within the prompt to reinforce the LLM's adherence to formatting specifications, thereby counteracting potential context forgetting (memory decay) associated with long input sequences.
3. **Processing Suggestions:** We incorporated expert-derived heuristics in natural language to aid the LLM. These suggestions were developed through a pre-examination of representative literature by domain experts, summarizing common textual patterns and embedding essential chemical knowledge to facilitate the correct interpretation of complex chemical narratives.

4. Processing Cases (Few-Shot Learning): To facilitate in-context learning, we included a set of few-shot examples consisting of raw input texts paired with their corresponding, manually verified structured outputs. This module allows the LLM to learn specific input-output mappings and formatting requirements by example.

To resolve complex co-reference relationships, the pipeline processes sub-headings and main text segments as distinct but related inputs, enabling the model to associate compound references within the synthetic procedures with their corresponding specific compounds defined in the sub-headings. Full details of the prompt templates and specific examples are provided in Supplementary Note 1.2.

Data standardization design

The standardization process was carried out in three consecutive steps. In step (1), the chemical compound dataset was created using compound names and their respective CAS numbers extracted from the Organic Syntheses³⁰ collection. Next, in step (2), compound identifiers were primarily obtained through CAS numbers, followed by additional identification using PubChem⁴⁰ nomenclature and OPSIN⁴¹ systematic name parsing for unprocessed entries. In step (3), molecular name vectorization was conducted to produce a comprehensive dataset, which allowed for precise matching of mined molecular names. For unit standardization, temporal values were consistently converted into minutes, with particular attention given to unconventional temporal expressions, such as “overnight”. Similarly, material quantities were standardized by converting all measurements into uniform units of grams and moles. Full details regarding the prompt engineering strategies and templates are provided in Supplementary Note 1.1.

Design of SynChat

The reaction database is architected into two distinct components to optimize retrieval efficiency and information density: a vectorized query database for semantic search and a supplementary metadata repository for detailed experimental context.

(1) Vectorization and Indexing: Key retrieval fields—including product SMILES, ReactionSMILES, chemical nomenclature, reaction types, and article titles—were vectorized using the all-MiniLM-L6-v2 model, a Sentence-BERT (SBERT) framework. This model transforms input text into 384-dimensional dense vector embeddings without dimensionality reduction. These embeddings were indexed and managed using ChromaDB⁵³, an open-source vector database, to facilitate high-speed semantic similarity search. The specific distribution of these embedding vectors is visualized in Supplementary Note 5.

(2) Data Retrieval and Generation: Detailed experimental parameters, including full synthetic procedures, yields, reagent quantities, reaction conditions, and citation metadata, are stored in the supplementary database in a JSON file. Upon receiving a user query, the system executes a retrieval process that identifies relevant entries based on vector similarity (semantic match) and exact character matching within the ChromaDB index. The system then retrieves the corresponding comprehensive procedural details from the supplementary database.

Finally, the retrieved context—combining the concise query features and the full textual descriptions—is concatenated and fed into the Large Language Model (LLM). This Retrieval-Augmented Generation (RAG) approach enables the LLM to generate tailored, evidence-based responses to user inquiries regarding specific reaction conditions and procedures.

Data availability

The mined data, code, and evaluation datasets generated in this study have been deposited in the Zenodo database under accession code (<https://doi.org/10.5281/zenodo.18523416>).

The raw textual data utilized in this study are available under restricted access for copyright and licensing restrictions; access can be obtained by subscription to the original publishers. The processed evaluation data are available at the GitHub repository (<https://github.com/DeepSynthesis/ReactionSeek.git>) under the file path `examples/OrganicSyntheses/reaction_extract/evaluate_raw.json`. The literature list used in this study is available in the GitHub repository under the file path `examples/OrganicSyntheses/OrgS_cite.csv`. The Source Data underlying the main text figures are provided in the cited Zenodo repository⁵⁴.

Code availability

The source code developed for this study, including all scripts and algorithms, has been deposited in a public GitHub repository <https://github.com/DeepSynthesis/ReactionSeek.git>.⁵⁴ All code is available under the MIT License. The primary results reported in this work were generated using the GLM-4 model. The vision language model for image recognition tasks was based on GLM-4V. A comprehensive list of all large language models used for the multi-model evaluation is provided in the main text. Our framework is designed to be model-agnostic. The choice of specific models for testing was to benchmark performance and guide users in model selection. The provided code-base includes documentation to assist users in configuring and integrating their models, including proprietary models requiring user-provided API keys or other services compatible with the OpenAI API format. For the chemical structure recognition task, the InDraw³⁴ OCSR tool was employed. Users may substitute this with any alternative chemical structure recognition tool suitable for their requirements.

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Author contributions

S.L. and Q.Y. conceived and directed the project. J.L. designed the research workflow and developed the programs and prompt engineering for the data mining framework. M.L. built and deployed SynChat. All authors contributed to the writing and revision of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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